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COMPUTER SCIENCE DEPARTMENT

REPORT OF FEDERATED PROJECT GLSI2 LCS2

TITLE

Design and Implementation of a Web Based Attendance Management System

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Summary Sheet

As part of the computerized management of internships and the archiving of reports, we request that you provide the following information:

- Training Program : LCS2_GLSI2
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- Internship Supervisor (Name and Surname) : MOUSSA Faouzi
- Keywords Characterizing Your Report (4 to 5 words maximum): Attendance Management, Web Application, Automation, Tunisian Universities, Scrum

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- Yassine Gharbi
- Wassim Mechergui
- Mohamed Taoufik Houria
- Ahmed Ben Haj Amor,

students in GLSI , pledge to abide by this charter.

Done at Tunis, on March 27, 2025.

Signatures:

- Yassine Gharbi :
- Wassim Mechergui :
- Mohamed Taoufik Houria :
- Ahmed Ben Haj Amor :

Synthèse du rapport en français

Introduction

Dans le cadre du module "Projet Fédéré" de l'année académique 2024-2025 à la Faculté des Sciences de Tunis (Université de Tunis El Manar), notre équipe, composée de Yassine Gharbi, Wassim Mechergui, Mohamed Taoufik Houria et Ahmed Ben Haj Amor, a entrepris la conception et le développement d'une application web intitulée SAGA (Système Automatisé de Gestion des Absences). Ce projet, encadré par M. Faouzi Moussa et réalisé au sein du département informatique (FST - DSI), s'inscrit dans un effort pour moderniser la gestion des présences dans les universités tunisiennes. Les systèmes traditionnels, tels que les feuilles de présence manuscrites ou les fichiers Excel, sont souvent inefficaces, sujets aux erreurs humaines et vulnérables aux manipulations par les étudiants. Ces lacunes entraînent une charge administrative importante pour les professeurs et les administrateurs, ainsi qu'un manque de transparence dans le suivi des absences.

SAGA a été conçu comme une solution numérique visant à automatiser et optimiser ce processus pour trois principaux acteurs : les étudiants, les professeurs et les administrateurs. En offrant une plateforme intuitive et sécurisée, notre système répond aux besoins spécifiques du contexte éducatif tunisien tout en s'appuyant sur des technologies modernes telles que React.js, Express.js, Node.js et MySQL. Cette synthèse présente les grandes lignes du projet, depuis son cadre initial jusqu'à sa réalisation, en passant par les choix méthodologiques et techniques adoptés.

Cadre et problématique du projet

Le projet s'est déroulé entre le 18 février et le 26 mars 2025, dans le cadre d'une évaluation académique, mais avec une ambition plus large : proposer une solution viable pour une application réelle dans les universités tunisiennes. La gestion traditionnelle des présences pose plusieurs problèmes. Tout d'abord, la prise de présence manuelle est chronophage et prone aux erreurs, notamment dans la transcription des données. Ensuite, les systèmes basés sur Excel, bien que numériques, manquent de fonctionnalités avancées comme la validation automatique des justifications ou la génération de rapports statistiques. Enfin, ces méthodes permettent aux étudiants d'exploiter des failles, par exemple en falsifiant des présences ou en retardant la soumission de justifications, ce qui compromet l'équité et la discipline au sein des établissements.

Face à ces défis, SAGA vise à offrir une alternative digitale qui automatise le processus de gestion des présences tout en améliorant l'accessibilité et la fiabilité des données. Les objectifs principaux incluent : l'enregistrement automatisé des présences, la possibilité pour les étudiants de soumettre des justifications d'absence en ligne, la simplification du travail des professeurs pour marquer les présences, et la mise à disposition d'un tableau de bord administratif pour valider les justifications et produire des rapports détaillés. Cette solution vise à réduire les erreurs, alléger la charge administrative et encourager une responsabilisation accrue des étudiants.

Méthodologie et organisation

Pour mener à bien ce projet, nous avons adopté la méthodologie agile Scrum, qui favorise la collaboration, la flexibilité et un développement itératif. Le travail a été organisé en quatre sprints de 8 jours chacun, permettant une progression structurée et adaptable. Les rôles au sein de l'équipe ont été clairement définis : Yassine Gharbi a agi en tant que Product Owner, responsable de la liaison avec les parties prenantes et de la définition des priorités ; Ahmed Ben Haj Amor a endossé le rôle de Scrum Master, facilitant la communication et la résolution des obstacles ; tandis que l'ensemble de l'équipe, incluant Wassim Mechergui et Mohamed Taoufik Houria, a contribué au développement en tant que développeurs.

Le processus Scrum incluait des réunions régulières : des *Daily Scrum* de 15 minutes pour synchroniser les activités quotidiennes, des *Sprint Reviews* de 4 heures pour évaluer les résultats à la fin de chaque sprint, et des *Sprint Retrospectives* de 3 heures pour analyser les points forts et les axes d'amélioration. Cette approche a permis une gestion efficace du projet, avec des ajustements constants en fonction des retours et des besoins émergents.

Le backlog produit a été structuré autour d'*epics* et d'*user stories*, définissant les fonctionnalités clés pour chaque acteur. Par exemple, les étudiants peuvent consulter leurs statistiques de présence et soumettre des justifications, les professeurs marquent les présences rapidement, et les administrateurs valident les demandes via une interface dédiée. La conception a été soutenue par l'utilisation du langage de modélisation UML (Unified Modeling Language), avec des diagrammes de cas d'utilisation et de classes pour visualiser l'architecture et les interactions du système.

Conception et développement

La phase de conception a reposé sur une architecture en trois couches : la couche de présentation (frontend) développée avec React.js, la couche applicative (backend) basée sur

Node.js et Express.js, et la couche de données gérée par MySQL. Cette séparation garantit une scalabilité et une maintenance facilitées. Les diagrammes UML ont permis de modéliser les relations entre les entités (étudiants, professeurs, administrateurs) et les fonctionnalités principales, comme l'authentification, la gestion des présences et la validation des justifications.

Le développement s'est déroulé en quatre sprints :

1. **Sprint 1 (18-26 février)** : Conception des maquettes avec Figma pour les pages de connexion, de suivi des présences, de soumission de justifications et de statistiques.
2. **Sprint 2 (27 février-7 mars)** : Implémentation du frontend avec React.js et JSX, utilisant des composants prédéfinis pour accélérer le développement.
3. **Sprint 3 (8-16 mars)** : Développement des API backend avec Express.js et Passport.js pour l'authentification, ainsi que des endpoints pour gérer les présences et les justifications.
4. **Sprint 4 (17-26 mars)** : Intégration du frontend et du backend pour assurer une communication fluide entre les couches.

Les outils utilisés incluent Visual Studio Code comme éditeur principal, Git et GitHub pour le contrôle de version, Thunder Client pour tester les API, et StarUML pour les diagrammes UML. Les technologies clés comprennent JavaScript, CSS, React, Node.js, Express.js et MySQL, offrant une combinaison de performance, flexibilité et robustesse.

Fonctionnalités et description de l'application

SAGA propose une interface conviviale adaptée à chaque acteur :

- **Page de connexion** : Une interface simple permettant aux utilisateurs (étudiants, professeurs, administrateurs) de se connecter avec leurs identifiants universitaires.
- **Pages étudiants** : Les étudiants peuvent consulter leurs taux de présence sous forme de graphiques et soumettre des justifications avec pièces jointes.
- **Pages professeurs** : Les professeurs accèdent à leur emploi du temps, marquent les présences et soumettent des justifications si nécessaire.
- **Pages administrateurs** : Un tableau de bord affiche les justifications en attente, avec des options pour les accepter ou les rejeter.

L'application utilise une base de données MySQL pour stocker les données de présence, les justificatifs et les informations des utilisateurs, tandis que Passport.js garantit une

authentification sécurisée. Les statistiques, comme le taux de présence ou l'impact des justifications, sont générées dynamiquement à partir des données collectées.

Conclusion et perspectives

SAGA représente une réponse concrète aux défis de la gestion des présences dans les universités tunisiennes. En automatisant les processus, elle réduit les erreurs humaines, améliore l'efficacité administrative et renforce la transparence. Sur le plan technique, ce projet nous a permis de maîtriser le développement full-stack et de travailler avec des outils modernes, tandis que sur le plan professionnel, il nous a sensibilisés aux besoins réels du système éducatif local.

À l'avenir, SAGA pourrait être enrichi avec des fonctionnalités supplémentaires, comme des notifications par email ou SMS, une intégration avec des systèmes universitaires existants, ou une application mobile pour une accessibilité accrue. Son déploiement à plus grande échelle nécessiterait des tests approfondis et une adaptation aux spécificités de chaque établissement. En somme, ce projet constitue une base solide pour une solution durable et évolutive dans le domaine de la gestion académique.

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General Introduction

In the modern educational landscape, efficient attendance management remains a significant challenge for universities. Traditional methods such as manual attendance sheets and Excel-based tracking systems are prone to errors, time-consuming, and inefficient. This often leads to students exploiting the system, resulting in an increase in unjustified absences and complications for professors and administrators when managing attendance records.

To address these issues, we developed ***SAGA*** *(Système Automatisé de Gestion des Absences), an automated absence management system tailored to the needs of* ***Tunisian universities.*** *SAGA offers a comprehensive solution that streamlines the attendance process for students, professors, and administrators.*

Our system enables students to track their attendance records and submit absence justifications directly through the platform. Professors benefit from a simplified attendance marking process, while administrators can efficiently validate justifications and generate attendance reports. By leveraging technologies such as ***React.js****,* ***Express.js****,* ***MySQL****, and* ***Passport.js****, SAGA ensures a secure, reliable, and user-friendly experience.*

Through automation and improved accessibility, SAGA not only reduces the administrative burden but also encourages students to maintain consistent attendance, fostering a more accountable learning environment. Our solution aligns with the Tunisian educational system's specific needs, offering a localized and effective approach to absence management.

CHAPTER 1: PROJECT PRESENTATION

1.1 Project Framework

This project was carried out as part of a university project at the Faculty of Science of Tunis. It was conducted within the context of the "Projet Fédéré" module and supervised by Mr. Faouzi MOUSSA.

The project spanned from February 18 to March 26. Although initially designed for academic evaluation, SAGA also holds strong potential for real-world application in Tunisian universities, providing a modern and effective solution for attendance management.

1.2 Problem Statement

The traditional attendance management methods at universities present several issues. Manual attendance sheets and Excel-based systems are slow, prone to human error, and susceptible to manipulation. Additionally, these methods create unnecessary administrative burdens when students need to submit justifications for their absences, causing delays and mismanagement in the approval process.

This inefficiency often results in inaccurate attendance records and increased workload for professors and administrators. Furthermore, students may exploit gaps in the system to manipulate attendance outcomes, undermining fairness and accountability.

To solve these challenges, we created SAGA, a digital platform designed to automate and improve the entire attendance management process. With features that allow students to submit justifications digitally and administrators to manage records efficiently, SAGA minimizes errors and ensures a transparent, streamlined process for all stakeholders.

1.3 Methodological Choices

1.4.1 Approach and Follow-Up

To ensure effective project development, we adopted the **Scrum** methodology, an agile framework that promotes collaboration, flexibility, and iterative progress. Our team worked in sprints, conducting regular stand-up meetings to track progress and adjust priorities as needed.

Our SCRUM team is made up of :

- Product Owner - "**Yassine Gharbi**" : The product owner focuses on the business side of product development and spends the majority of time liaising with stakeholders and the team.
- Scrum Master - "**Ahmed Ben Haj Amor**" : scrum master role is to educate and coach teams about scrum theory and practice. Scrum masters have differing roles and responsibilities from traditional team leads or project managers. Some scrum master responsibilities include coaching, objective setting, problem solving, oversight, planning, backlog management, and communication facilitation.
- Developers Team - "**All the team**" : In scrum, the term *developer* or *team member* refers to anyone who plays a role in the development and support of the product and can include researchers, architects, designers, programmers, etc.

Workflow :

- Sprint Reviews: Once a sprint is completed, a meeting is held that lasts a maximum of 4 hours, with the participation of everyone involved (Scrum Master, the development team, the Product Owner).
- Sprint Retrospective: This is an internal meeting (i.e., Scrum team only) lasting a maximum of 3 hours. The purpose of this meeting is to reflect on the completed sprint to determine what went well and what needs improvement.
- Daily Scrum Meeting: Also known as a "stand-up meeting", this is a daily 15-minute meeting that allows the team to synchronize their activities and create a plan for the next 24 hours. Each member of the development team must answer the following three questions:
    1. What did they accomplish yesterday?
    2. What will they accomplish today?
    3. What obstacles are holding them back?

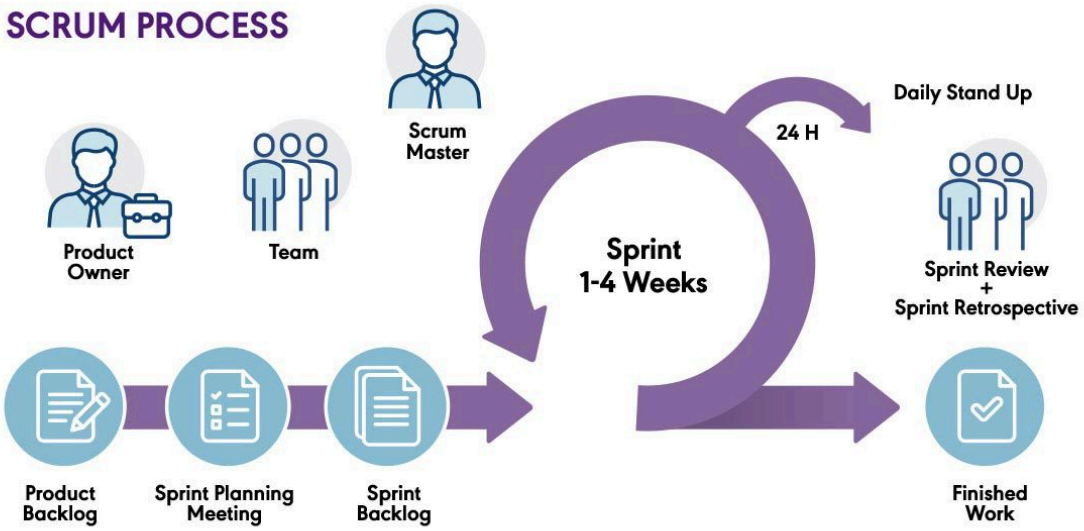


Figure 1 :Scrum Process

1.4.2 Modeling Language:

For the design phase, we utilized **UML (Unified Modeling Language)** to create clear and structured diagrams that illustrate the system's architecture, data flow, and key functionalities. This approach ensured a well-organized development process and facilitated effective communication among team members.

CHAPTER 2: REQUIREMENTS SPECIFICATION

2.1 Requirements Description

SAGA (Automated Attendance Management System) is designed to streamline and digitalize the attendance tracking process within higher education institutions. The system's primary objectives include:

- **Automated Attendance Registration:** Replace manual attendance sheets with a digital solution that allows real-time tracking.

- **Absence Reporting and Justification:** Enable students to report and justify their absences easily.

- **Simplified Process for Professors:** Provide an intuitive interface for professors to mark attendance and manage student records.

- **Administrative Oversight:** Offer a centralized dashboard for administrators to validate justifications and generate comprehensive attendance reports.

These requirements ensure that the system not only improves efficiency but also reduces administrative burdens while aligning with the specific needs of the Tunisian educational environment.

2.2 Stakeholder Identification

The following key actors are involved in the SAGA application:

- **Students:** They will be able to access their attendance records, submit absence justifications, and receive notifications regarding their attendance status.
- **Professors:** Tasked with digital attendance marking, they are responsible for verifying and managing the attendance data of their classes.
- **Administrators:** They oversee the overall attendance data, validate absence justifications submitted by both students and professors, and generate the necessary reports for university records.

2.3 Product Backlog

- User Stories: A user story is a short, simple description of a feature told from the perspective of the person who desires the new capability, usually a user or customer of the system (https://www.mountaingoatsoftware.com/agile/user-stories)
- Product Backlog : A product backlog is a prioritized list of work for the development team that is derived from the product roadmap and its requirements (https://www.atlassian.com/agile/scrum/backlogs)

Epic	User story
Epic 1: User Authentication	**1.1** ***As a*** *student*, **I want to** register using my university Email and password **so that** I can access my account.
	1.2 ***As a*** *professor*, **I want to** log in using my university Email and password **so that** I can access my account.
	1.3 ***As an*** *administrator*, **I want to** log in with my credentials **so that** I can validate justifications
Epic 2: Professor Features	**2.1** ***As a*** *professor*, **I want to** mark attendance quickly **so that** I can record who is present in my class.
	2.2 ***As a*** *professor*, **I want to** submit a justification for an absence **so that** my absence can be reviewed by the administrator.
Epic 3: Student Features	**3.1** ***As a*** *student*, **I want to** view my attendance records **so that** I can track my presence in each class.
	3.2 ***As a*** *student*, **I want to** submit a justification for an absence **so that** my absence can be reviewed by the administrator.
Epic 4: Administrator Features	**4.1** ***As an*** *administrator*, **I want to** view submitted justifications **so that** I can review and validate them.

Table 1 : Product backlog table

- Sprints: A sprint is a short, time-boxed period when a scrum team works to complete a set amount of work (https://www.atlassian.com/agile/scrum/backlogs)

  we decided to divide our workflow in 4 sprints each one with 8 days :

Sprint	Period	Tasks
Sprint 1	From 18 february to 26 february	1.1 Design the sign-in page mockup using Figma
		1.2 Design the “attendance tracking page” using Figma
		1.3 Design the “send justification page” using Figma
		1.4 Design the “Attendance Statistics” page using Figma
		1.5 Design the “Pending Justifications” page using Figma
Sprint 2	From 27 february to 7 march	Implement the sign-in page using React and JSX syntax
		Implement the attendance tracking page using React JSX syntax **Task Details** : You can utilize prebuilt components and external libraries to implement some elements in the page (such as the schedule).
		Implement the “send justification” component using React JSX syntax **Task Details** : You can utilize prebuilt components and external libraries to implement some elements in the page (such as the attachment upload field).
		Implement the “Attendance

		Statistics” page using React JSX syntax **Task Details** : You can utilize prebuilt components and external libraries to implement some charts in the page
		Implement the “Pending Justifications” page using React JSX syntax
Sprint 3	From 8 march to 16 march	Implement backend API for sign-in feature. **Task Details** : Use Passport Js for authentication
		Implement backend APIs to handle the attendance marking procedure **Task Details** : The APIs we need are: 1/ fill the schedule of the professor 2/ get list of students of a particular class (or groupe) 3/ mark attendance for each student
		Implement backend APIs to handle the justification sending procedure
		Implement backend APIs to provide the data we need to fill the charts **Task Details** : The APIs we need are: 1/ Get the presence ratio 2/ Get the data to calculate and plot the attendance coherence over time 3/ Retrieve the impact of justifications, including the number of accepted justifications, rejected justifications, and those still

		awaiting review by the administration.
		Implement backend APIs to handle justifications **Task Details** : The APIs we need are: 1/ Get all pending (not reviewed) justifications 2/ Accept or reject a particular justification
Sprint 4	From 17 march to 26 march	Integrate backend APIs with the front-end

Table 2 : Sprints Backlog table

3.1 Class diagram

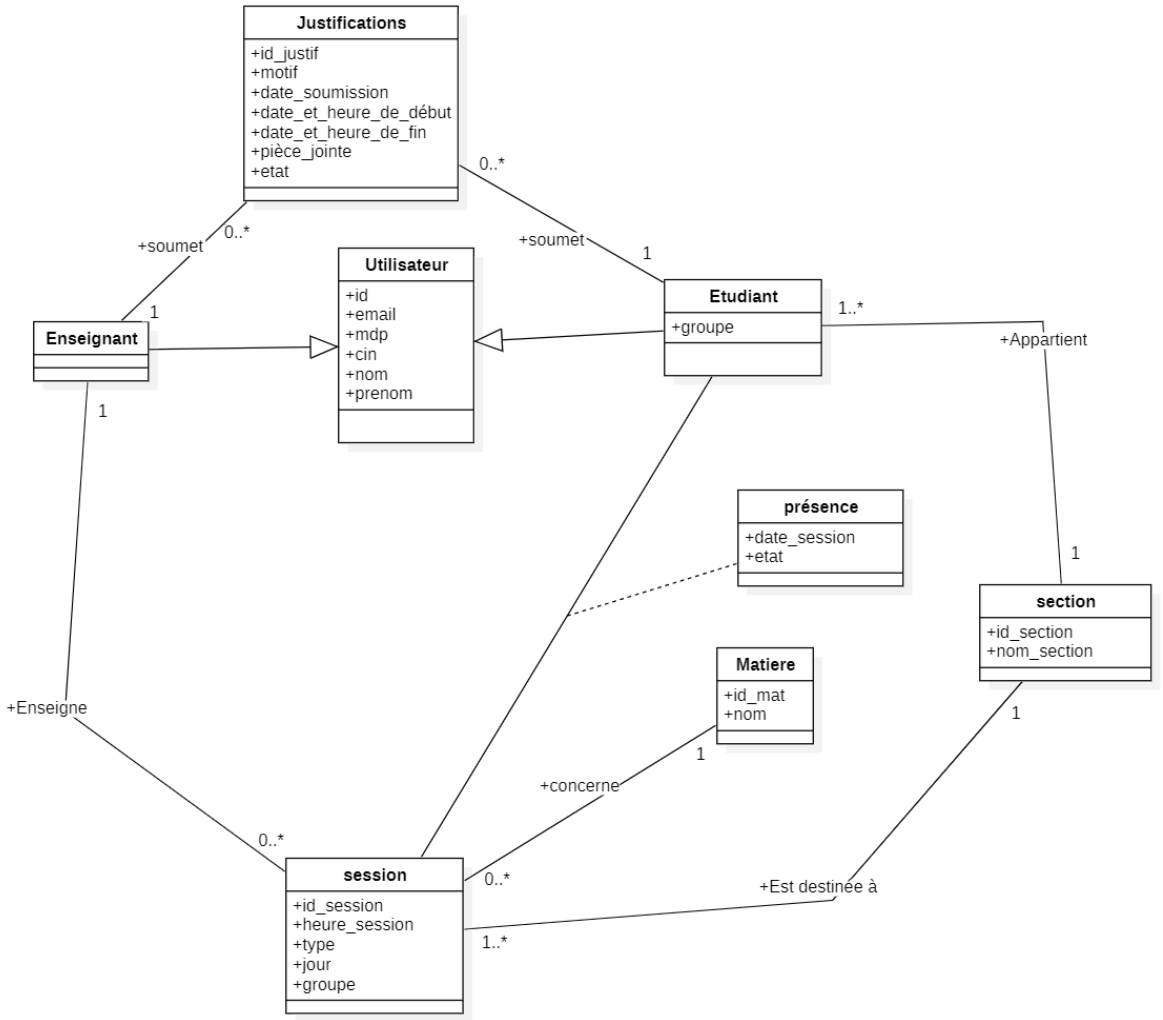


Figure 2 : Class diagram

2.3 Use case diagram

3.1.1 General use case diagram

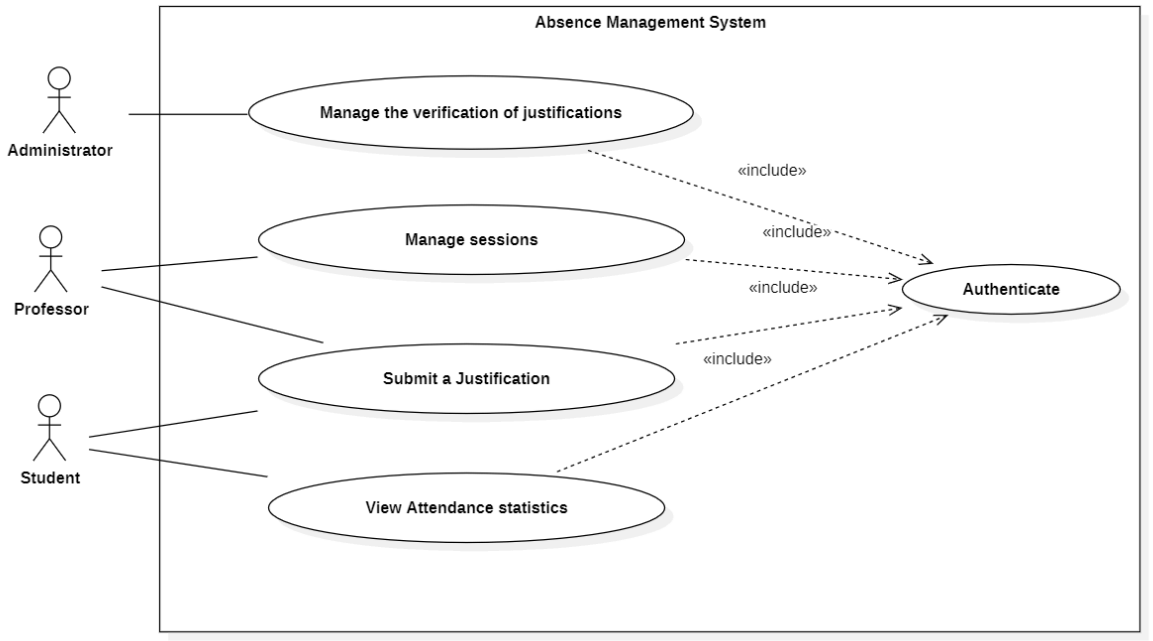


Figure 3 : General use case diagram

3.1.2 Detailed use case diagram

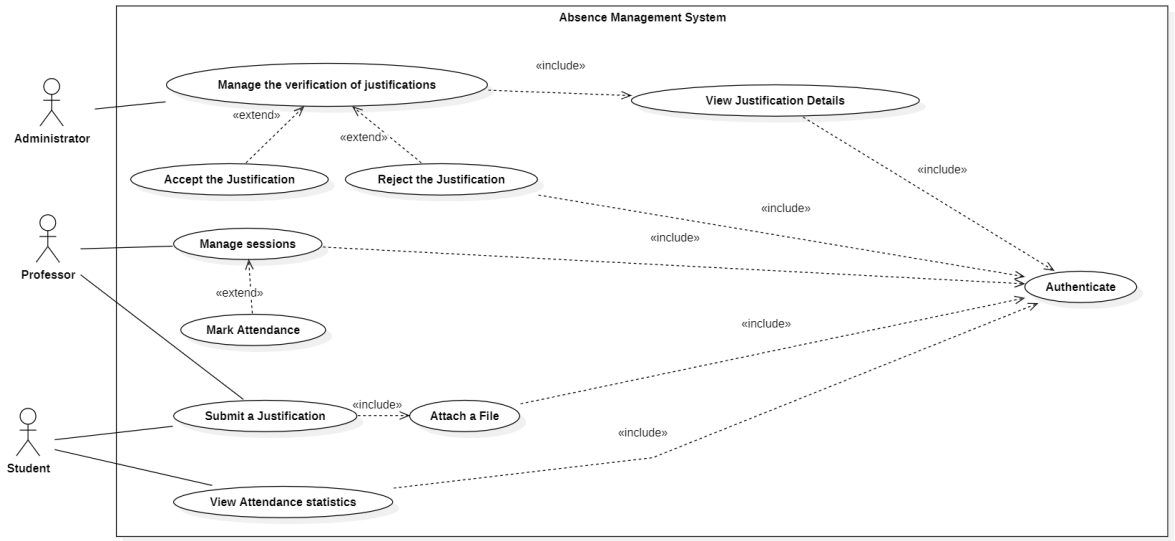


Figure 4 : Detailed use case diagram

3.1.3 Sequence diagram

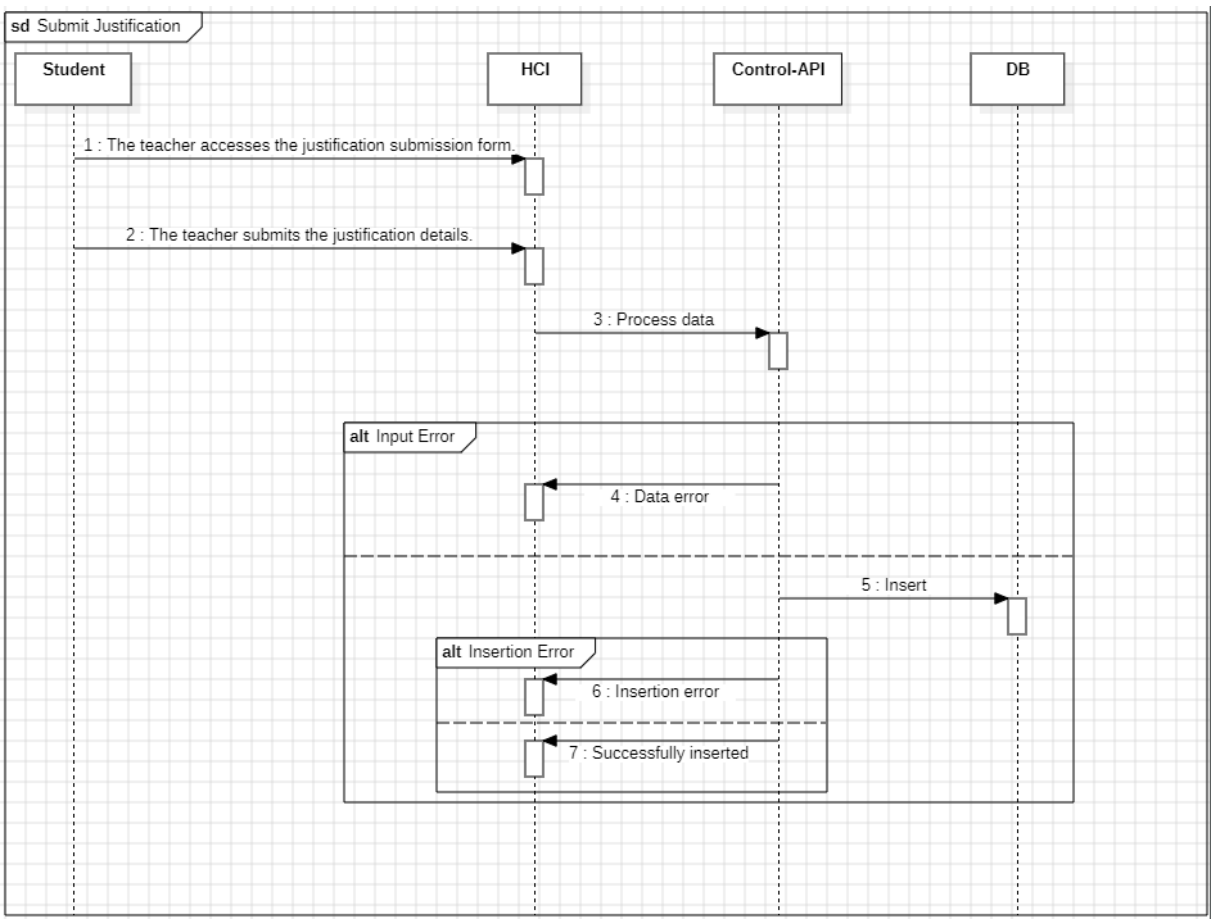


Figure 5 : Submit justifications sequence diagram

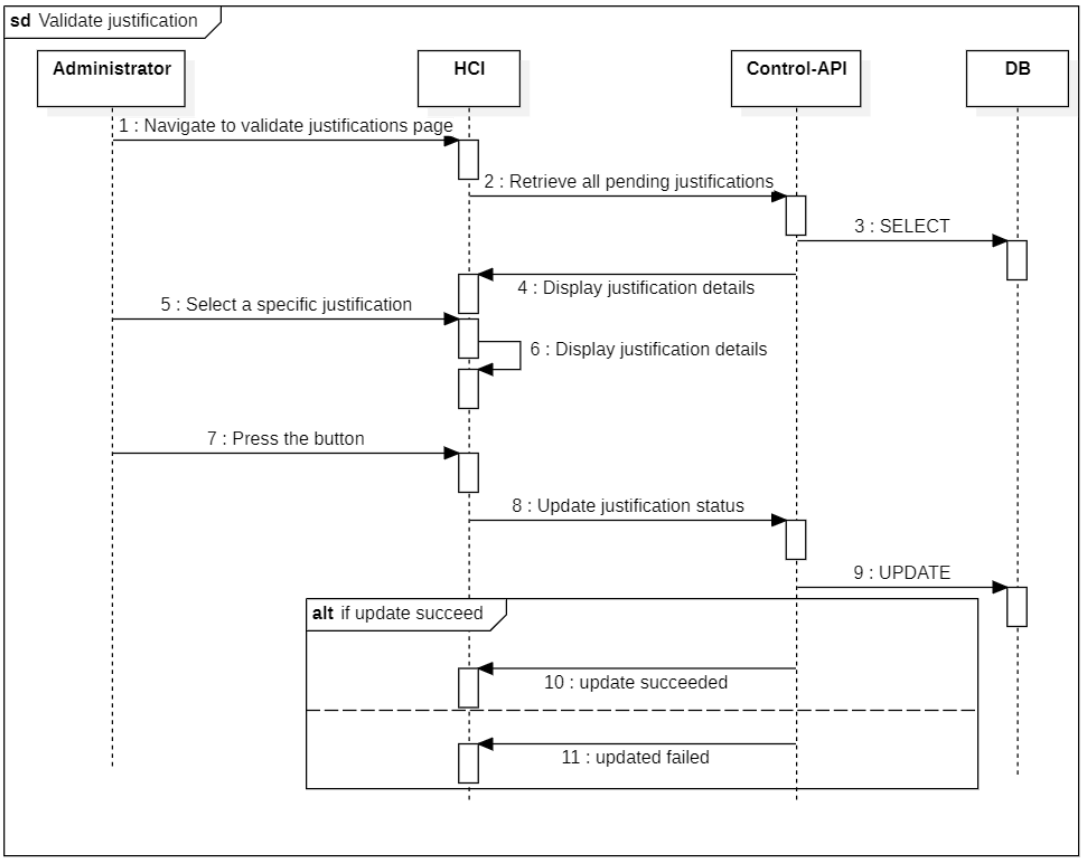


Figure 6 : Verify justifications sequence diagram

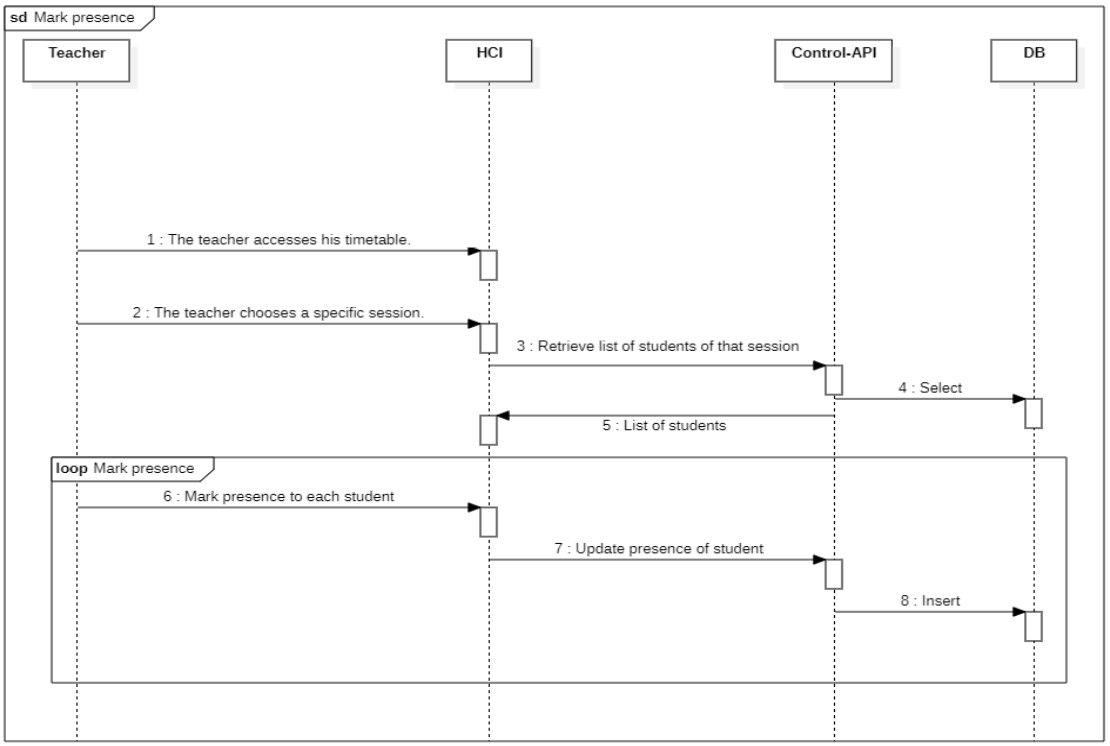


Figure 7 : Mark presence sequence diagram

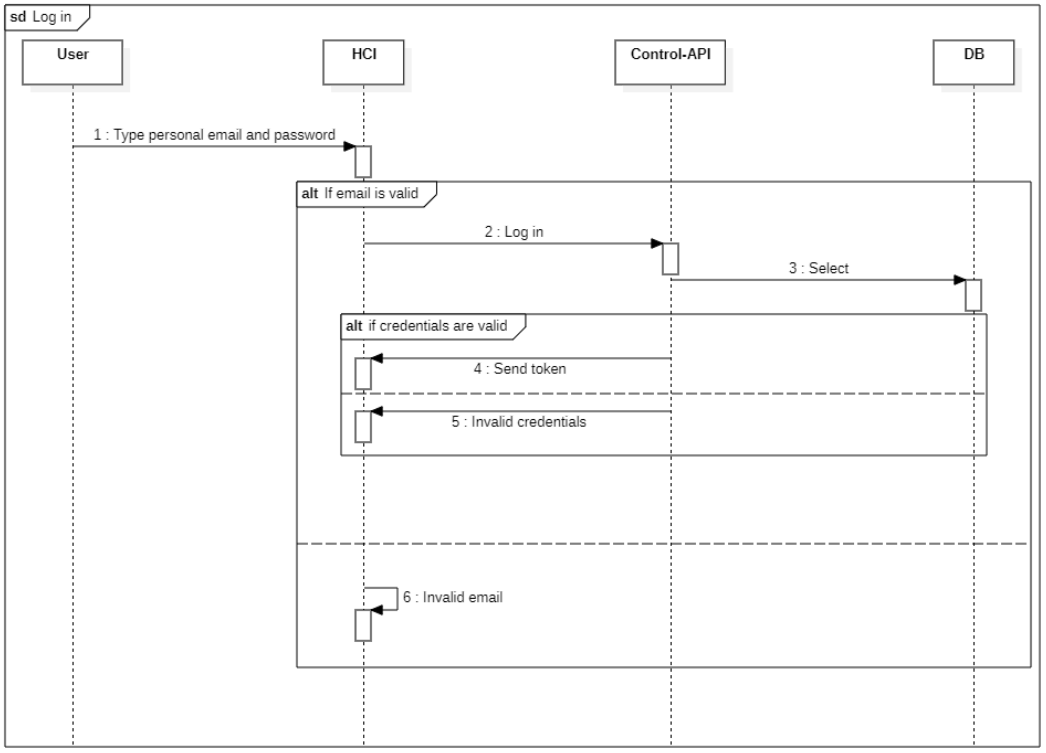


Figure 8 : Login sequence diagram

4.1 Software Environment

Software name	Logo	Usage
Visual Studio Code		Visual Studio Code features a lightning fast source code editor, perfect for day-to-day use. With support for hundreds of languages, VS Code helps you be instantly productive with syntax highlighting, bracket-matching, auto-indentation, box-selection, snippets, and more.
Git & GitHub	git GitHub	Git is a distributed version control system that tracks changes in code, allowing multiple developers to collaborate efficiently. GitHub is a cloud-based platform that provides hosting for Git repositories, enabling version control, collaboration, and project management.
Thunder Client		Thunder Client is a lightweight API testing extension for VS Code that allows developers to send HTTP requests and inspect responses. It provides an intuitive interface for testing RESTful APIs without needing a separate tool like Postman.
StarUML		StarUML is a software modeling tool that supports UML diagrams for designing and visualizing software architectures. It provides an intuitive interface for creating class diagrams, sequence

		diagrams, and other UML models to aid in system analysis and development.
Google Docs	Google Docs	Google Docs is a cloud-based word processor that allows users to create, edit, and collaborate on documents in real time. It offers features like automatic saving, version history, and seamless sharing across devices.

Table 3 : Software Environment table

4.2 Tools (Languages, Libraries …)

Tool	Logo	Usage
JavaScript	JS	JavaScript is a high-level, interpreted programming language used to create dynamic and interactive web content. It enables client-side functionality, such as animations, form validation, and API interactions .
Node.js	node JS	Node.js is a runtime environment that allows JavaScript to run on the server side, built on Chrome's V8 engine. It enables scalable, non-blocking, event-driven applications, making it ideal for building APIs, web servers, and real-time applications.
CSS	CSS 3	CSS (Cascading Style Sheets) is a stylesheet language used to describe the presentation and layout of web pages. It controls the design aspects such as colors, fonts, spacing, and positioning, enabling developers to create visually appealing and responsive web pages.

React		React is a JavaScript library for building user interfaces, primarily for single-page applications. It allows developers to create reusable UI components and manage state efficiently, enabling the development of dynamic, fast, and interactive web applications.
Express.js		Express.js is a minimal and flexible Node.js web application framework that simplifies building server-side applications. It provides a robust set of features for routing, middleware, and handling HTTP requests, making it easier to develop APIs and web applications.
MySQL		MySQL is an open-source relational database management system (RDBMS) that uses SQL (Structured Query Language) to manage and manipulate data. It is widely used for storing and retrieving data in applications, providing features like data integrity, security, and scalability.

Table 4 : Tools table

4.3 Application architecture

This is a **3-tier architecture** for a full-stack web app:

1. **Frontend (React) - Presentation Layer**: Manages UI and navigation (React Router) and communicates with the backend using Axios.

2. **Backend (Node.js + Express) - Application Layer**: Handles API routes, business logic, and interacts with the database via an ORM.

3. **Database (MySQL) - Data Layer**: Stores and retrieves persistent data.

The app follows a **3-tier model**, separating presentation, logic, and data layers for better scalability.

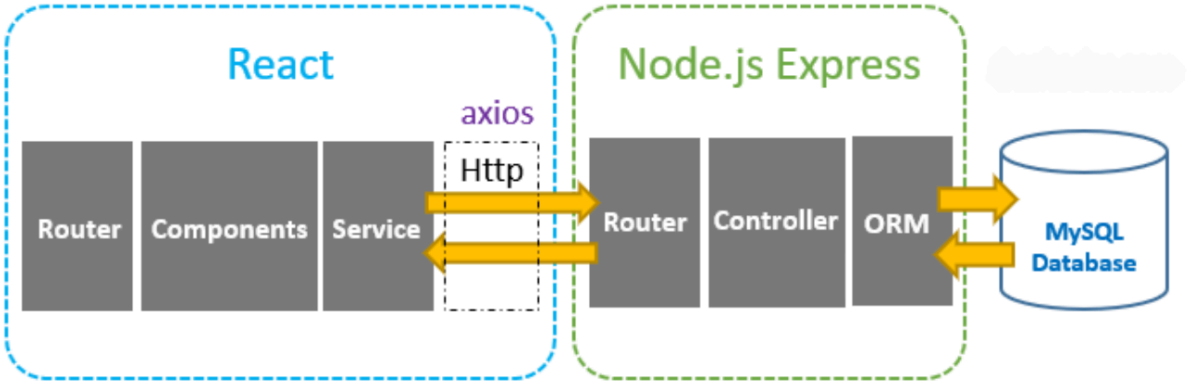
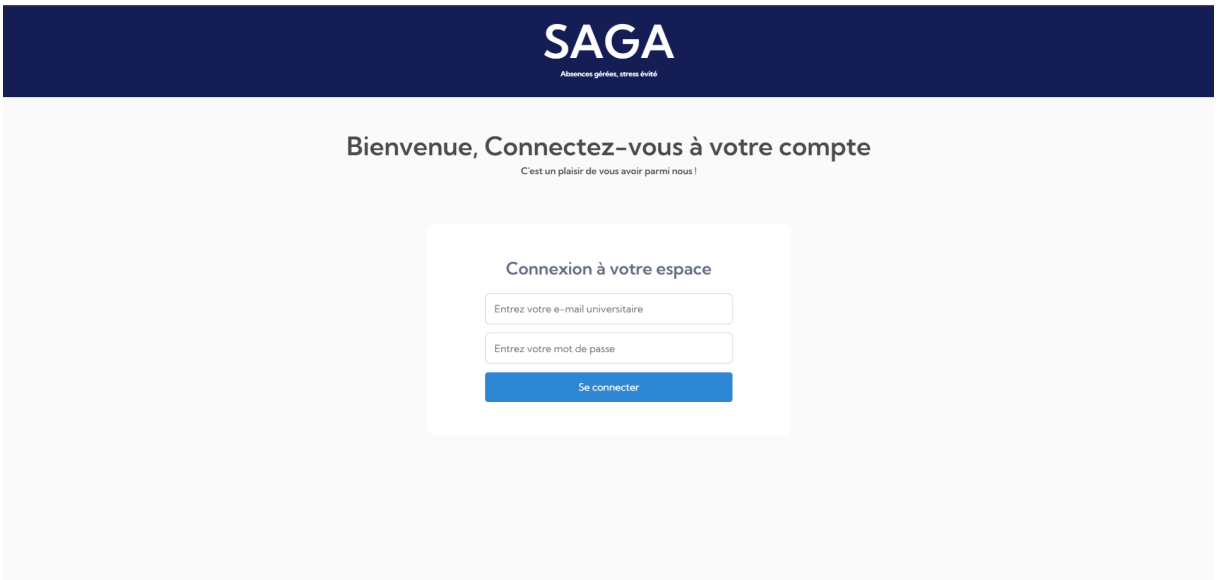


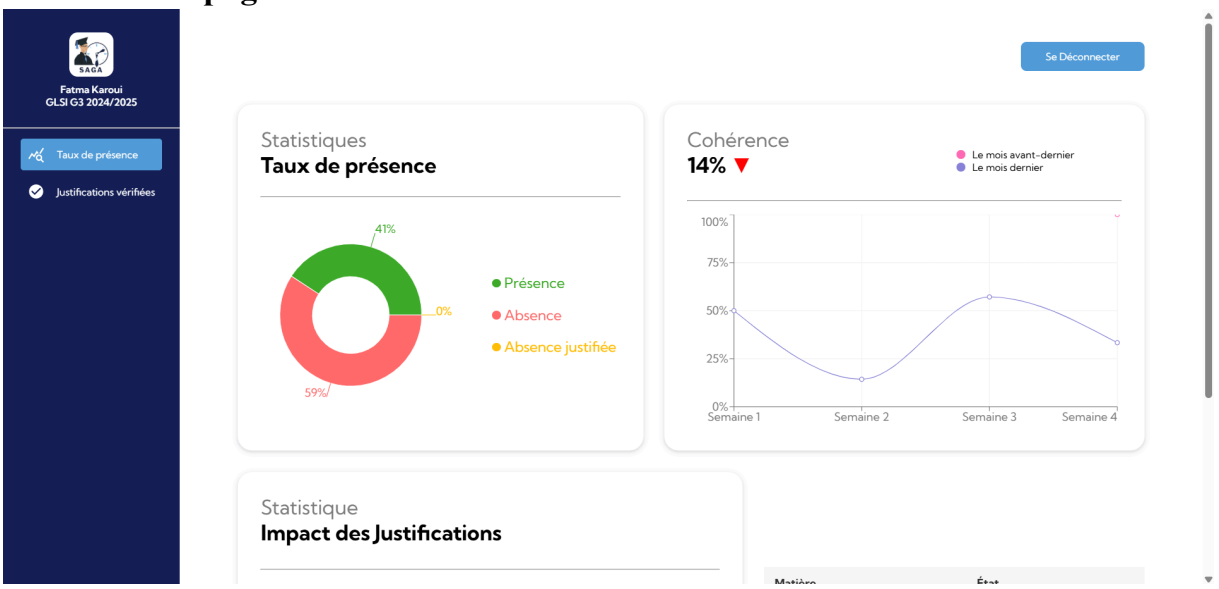
Figure 9 : Application architecture

4.4 Application description

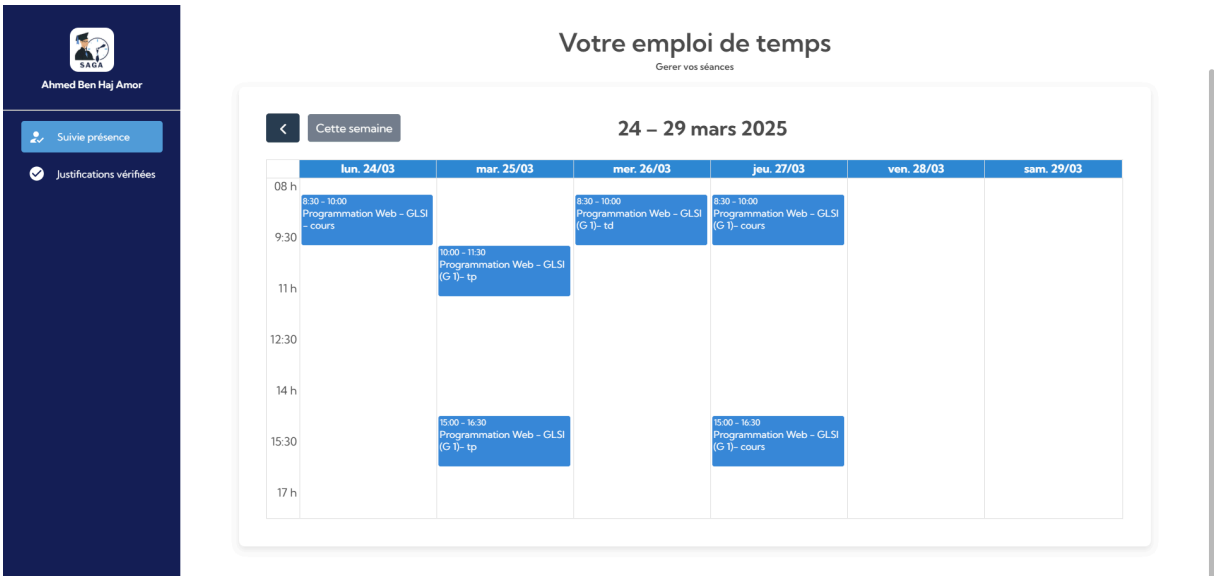
4.4.1 Log-in page

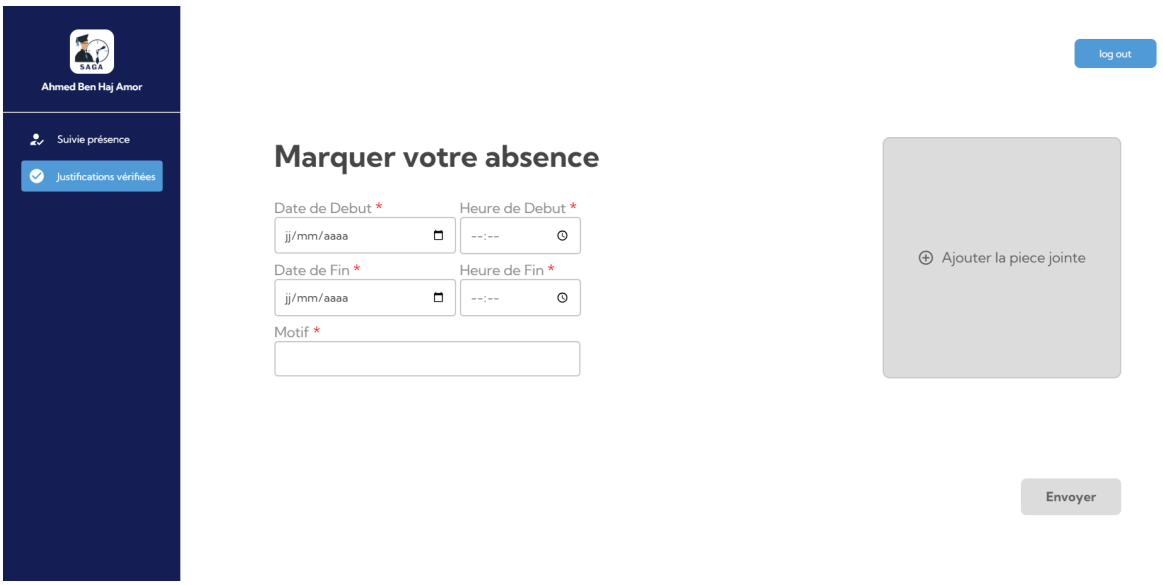


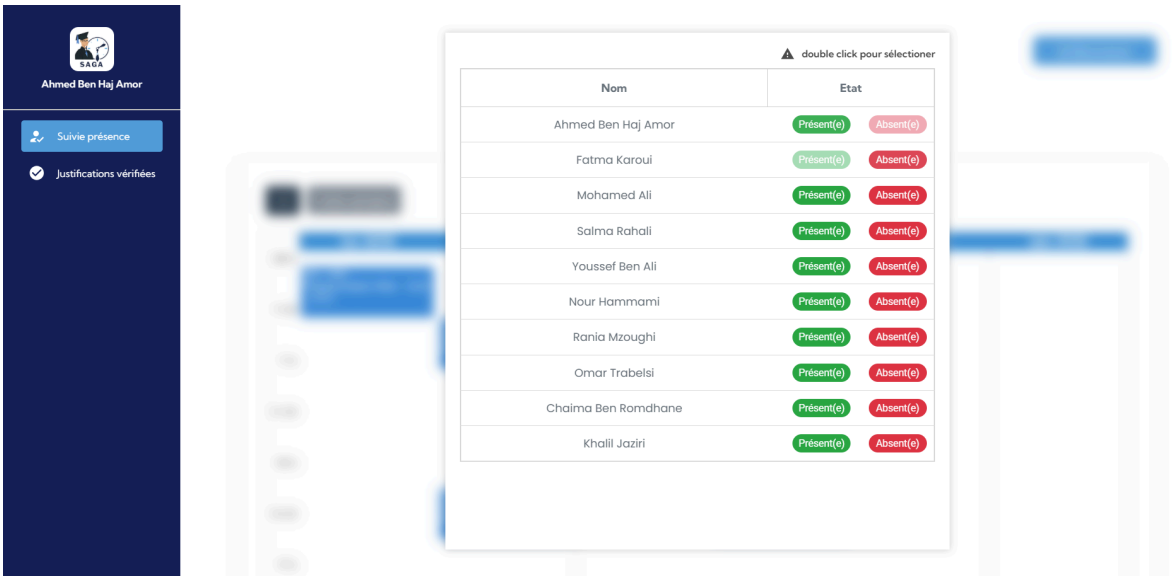
4.4.2 Student pages



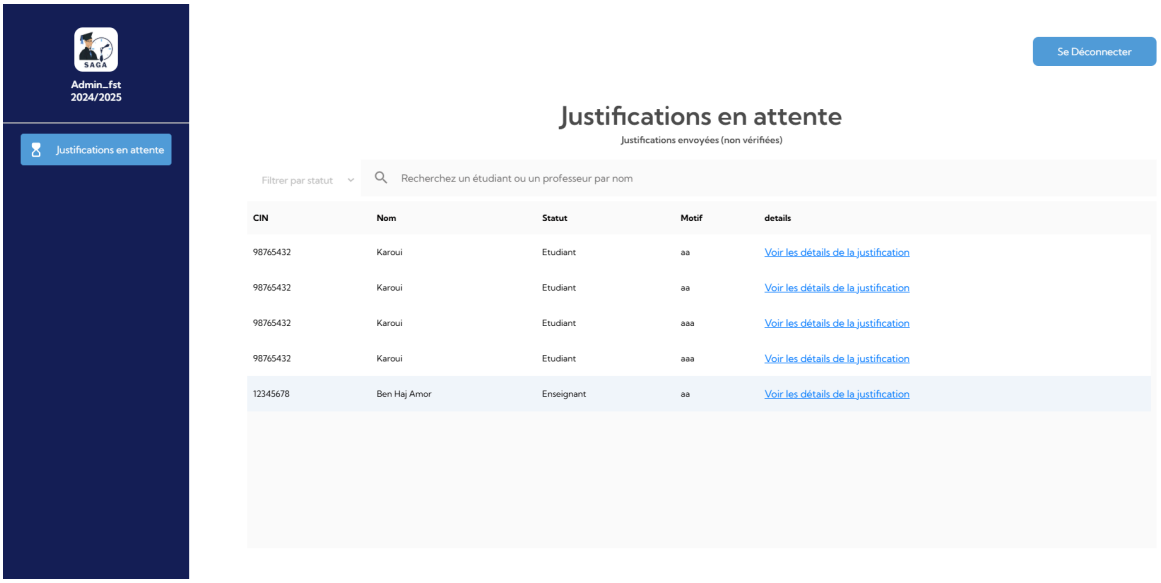
4.4.3 Professor pages

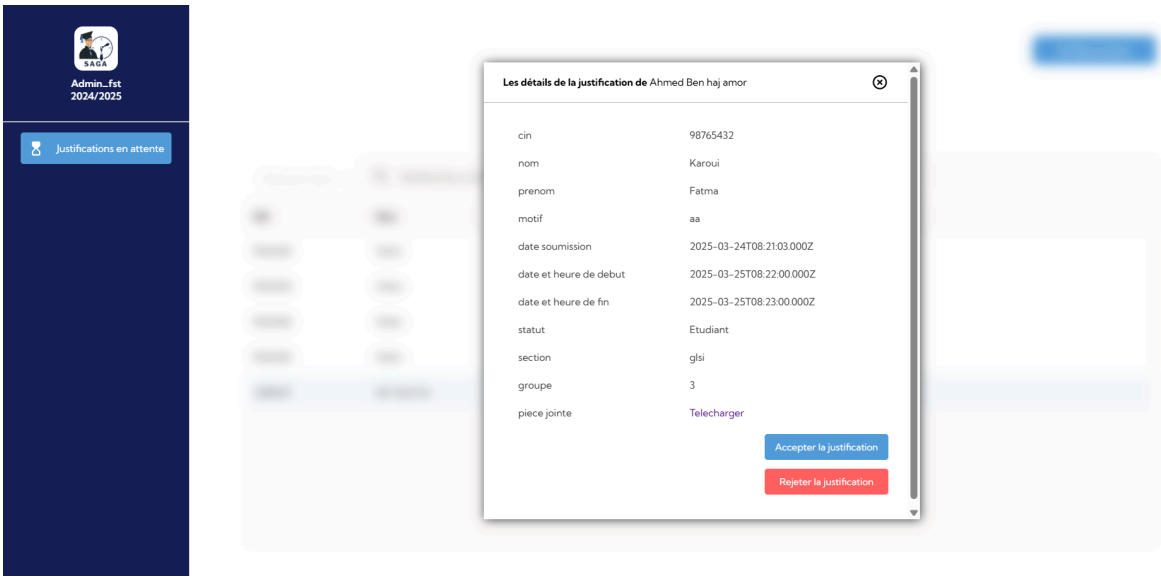






4.4.2 Administrator pages





General Conclusion and Perspectives

The SAGA (Automated Attendance Management System) project, undertaken as part of the "Federated Project" module at the Faculty of Sciences of Tunis during the 2024-2025 academic year, successfully delivered an innovative web-based solution to address the inefficiencies of traditional attendance management in Tunisian universities. Conventional methods, such as manual attendance sheets and Excel-based systems, were prone to errors, time-consuming, and vulnerable to manipulation, resulting in administrative burdens and a lack of transparency. Our team—Yassine Gharbi, Wassim Mechergui, Mohamed Taoufik Houria, and Ahmed Ben Haj Amor—developed SAGA to automate attendance tracking, streamline justification submissions, and enhance administrative oversight, under the supervision of Mr. Faouzi Moussa.

Spanning from February 18 to March 26, 2025, the project tackled key challenges by providing tailored functionalities for students, professors, and administrators. Students can now view their attendance records and submit justifications online, professors benefit from a simplified marking process, and administrators gain access to a centralized dashboard for validating submissions and generating reports. Leveraging a three-tier architecture—React.js for the frontend, Node.js with Express.js for the backend, and MySQL for data management—and adopting the Scrum methodology, we delivered a functional prototype within a month. This approach ensured iterative progress, effective collaboration, and adaptability to evolving requirements.

Technically, the project enhanced our skills in full-stack development, database management, and agile teamwork, while professionally, it deepened our understanding of the Tunisian educational system's needs. Despite its successes, SAGA has limitations, such as the lack of integration with existing university systems and a mobile interface. Looking ahead, potential improvements include real-time notifications, mobile app development, and customization for broader institutional use. SAGA lays a strong foundation for a scalable, impactful solution, with the potential to modernize attendance management beyond this academic exercise.

Bibliography

Websites visited

[1] Mountain Goat Software, 26/03/2025,
https://www.mountaingoatsoftware.com/agile/user-stories

[2] Atlassian, 26/03/2025, https://www.atlassian.com/agile/scrum/backlogs

[2] Wikipidia, 27/03/2025, https://en.wikipedia.org/wiki/Scrum_(software_development)

Summary

The SAGA project aimed to design and implement a web-based attendance management system to replace outdated manual and Excel-based methods in Tunisian universities. Developed by a team of four students at the Faculty of Sciences of Tunis, the system addresses inefficiencies, errors, and manipulation risks by automating attendance tracking and justification processes. Built using React.js, Node.js, Express.js, and MySQL, SAGA features a user-friendly interface for students to monitor attendance and submit justifications, for professors to mark attendance, and for administrators to oversee and validate data. The project, completed between February and March 2025 using the Scrum methodology, demonstrates a practical solution tailored to local educational needs, reducing administrative workload and improving accountability. Future enhancements could include mobile access and system integrations.

Keywords: Attendance Management, Web Application, Automation, Tunisian Universities, Scrum

Abstract

Efficient attendance management remains a challenge in higher education, particularly in Tunisian universities reliant on manual or Excel-based systems prone to errors and manipulation. This report details the design and implementation of SAGA, a web-based Automated Attendance Management System developed during the 2024-2025 academic year at the Faculty of Sciences of Tunis. Utilizing React.js, Node.js, Express.js, and MySQL within a three-tier architecture, SAGA automates attendance tracking, justification submissions, and reporting for students, professors, and administrators. Employing the Scrum methodology, the system was completed between February 18 and March 26, 2025, offering a scalable, user-centric solution that reduces administrative burdens and enhances transparency. While successful as a prototype, future work could integrate mobile functionality and university-wide systems to broaden its impact.

Keywords: Web-Based System, Attendance Automation, Higher Education, Agile Development, SAGA

ملخص

الفصل الأول: عرض المشروع

تم تنفيذ المشروع في الفترة من 18 فبراير إلى 26 مارس 2025 ضمن كلية العلوم بتونس. يهدف SAGA إلى حل مشاكل الأنظمة التقليدية مثل بطء التسجيل، الأخطاء البشرية، والعبء الإداري الثقيل الناتج عن تبرير الغيابات يدويًا. اعتمدنا منهجية Scrum الرشيقة لتطوير المشروع، مع تقسيم العمل إلى دورات (sprints) واجتماعات يومية ومراجعات دورية، مما سمح بتتبع التقدم وضبط الأولويات بفعالية. استخدمنا لغة النمذجة الموحدة (UML) لتصميم النظام وتوضيح هيكليته.

الفصل الثاني: مواصفات المتطلبات

حددنا متطلبات SAGA لتشمل تسجيل الحضور التلقائي، تقديم تبريرات الغياب بسهولة، تبسيط عملية التسجيل للأساتذة، وتوفير لوحة تحكم للإداريين للتحقق من البيانات وإعداد التقارير. الجهات المعنية تشمل الطلاب (لمتابعة حضورهم)، الأساتذة (لتسجيل الحضور)، والإداريين (للمصادقة وإدارة السجلات). تم إنشاء قائمة أولويات المنتج (Product Backlog) تحتوي على قصص المستخدمين، مقسمة إلى أربع دورات عمل: تصميم الواجهات، تطوير الواجهة الأمامية، تطوير الخلفية، والتكامل.

الفصل الثالث: التصميم

اعتمد تصميم SAGA على هيكلية ثلاثية الطبقات: الواجهة الأمامية (React.js)، الخلفية (Node.js وExpress.js)، وقاعدة البيانات (MySQL). تم إنشاء مخططات الفئات ومخططات حالات الاستخدام باستخدام UML لتوضيح العلاقات بين الكيانات (الطلاب، الأساتذة، الإداريون) والوظائف الأساسية مثل تسجيل الدخول، تتبع الحضور، وإدارة التبريرات.

الفصل الرابع: التنفيذ

استخدمنا بيئة برمجية تشمل Git، Visual Studio Code وThunder Client، GitHub، وStarUML، مع أدوات مثل Express.js، Node.js، React، CSS، JavaScript، وMySQL. تم تطوير النظام عبر أربع دورات: تصميم الواجهات باستخدام Figma، برمجة الواجهة الأمامية بـ React، تطوير واجهات برمجة التطبيقات (APIs) بـ Express وPassport.js للمصادقة، وأخيرًا دمج الواجهة مع الخلفية. يتضمن التطبيق صفحات لتسجيل الدخول، ومتابعة الحضور للطلاب، وتسجيل الحضور للأساتذة، وإدارة التبريرات للمسؤولين.

الخلاصة العامة والآفاق

نجح SAGA في تقديم حل رقمي يقلل من الأخطاء، يخفف العبء الإداري، ويعزز الشفافية في إدارة الحضور. المشروع عزز مهاراتنا التقنية في التطوير الكامل وإدارة قواعد البيانات، وزاد من فهمنا لاحتياجات النظام التعليمي التونسي. مع ذلك، يحتاج النظام إلى تكامل مع أنظمة جامعية أخرى وتطبيق محمول. في المستقبل، يمكن إضافة إشعارات فورية وتوسيع النظام ليشمل مؤسسات أخرى، مما يجعله حلاً مستدامًا وقابلاً للتطوير.

الكلمات المفتاحية: إدارة الحضور، تطبيق ويب، أتمتة، الجامعات التونسية، Scrum